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Quick introduction to Python

Sommario

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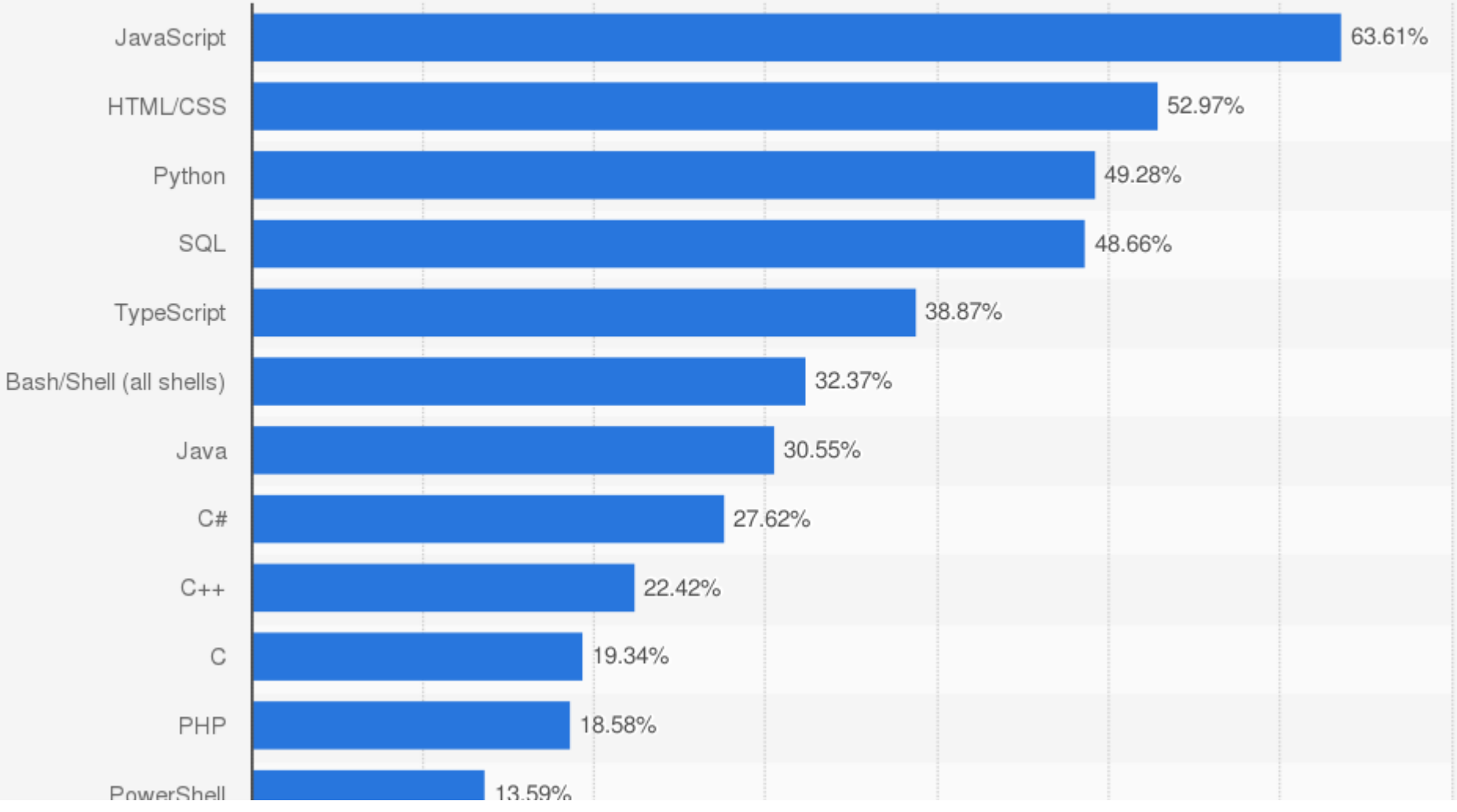
- TBD

Most used programming languages among developers worldwide as of 2023



Programming languages

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- Most programming languages are
  - General Purpose
- Selecting one can be difficult
  - Specifically for learning coding
- Practically different applicability
  - C → small programs/efficiency aka embedded systems like IoT
  - C++ → large programs where efficiency is a requirement
  - Python → when business speed matters more compared to execution speed

History

- Started by Guido Von Rossum as a ђobby
- Now widely spread
- Open Source ⟶ free
- Versatile
- Lot of additional libraries

Python today

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- Widely used for scientific computing & data analysis
- One of the most important languages for
  - Data science
  - Machine & deep learning
- Hundredths of additional libraries

Main Characteristics

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- High level language
- Easy “quick start”
- Often used as first programming language
- Object Oriented
- Extensible
- Interpreted language

Quick start

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- Interpreted language
- Namely, we can directly run the code we write
- You can open a shell and directly test this
- You can install python or
  - online environments

https://www.programiz.com/python-programming/online-compiler/

Try it yourself!

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- Open “programiz” web site and simply run the default code

https://www.programiz.com/python-programming/online-compiler/

Main elements

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- As other languages we have a **lexicon** and **syntax** rules
- Namely
  - Instructions
  - How to combine them
- Moreover
  - Do not forget **semantics**
  - What we write must also make sense

Main elements

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- Input & Output (I/O)
- Variables and costants
- Expressions
- Data structures
- Control flow
- Functions or subroutines

Comments

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- "Notes" that can be distributed along your code
- Human readable explanations
- They improve how to understand the code
- In Python everything after a "#" until a "newline"
  - Also text inside """ and """ is ignored and then used as comment

When a programmer first creates his code, only he and God know how it works, a few months down the line and only God knows

Brian W. Kernighan
C programming language developer

Try it yourself!

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- Add some “comments” to the default code
- Which character(s) can be used for comments?
  - %
  - //
  - #
  - /*

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Input & Output

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- "print()" instruction can be used to print on the screen
- We can also print different things separating them using a ','
  - A space is then used to separate them
- A newline is added at the end of the output

```
print("Hello world!")        # print a "string"
print(3)                     # print a "number"
print("My age is ", 57)      # combining multiple outputs
```

Input & Output

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- “input()” instruction can be used to read some “text” as input
- Wen issuing the input() an optional prompt message can be passed
- The input() stops until the enter key is pressed

```
name = input("What is your name? ")  # ask user
print("Hello", name)
```

Input & Output

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- “input()” always read “text”, namely a sequence of symbols
  - Even when a number is passed as input
- We need to convert other data…

```
age = input("What is your age? ") # ask user
print("Oh, you look younger than", age, "years")
print("I am", age + 10, "years old")

# the last line will give you a
# “TypeError: can only concatenate str (not "int") to str”
# error
```

Input & Output

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- "input()" always read "text", namely a sequence of symbols
  - Even when a number is passed as input
- We need to convert other data…

```
age = int(input("What is your age? "))  # ask user
print("Oh, you look younger than", age, "years")
print("I am", age + 10, "years old")

# now it works flawlessly
```

Variables & Constants

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- Python is a Typed Programming Language
  - Actually a strong dynamically typed language
- i.e. data feature a “type”
  - Integer number, floating point number, string…

Variables & Constants

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- How to write data ("simple" ones), namely constant values
  - 42 → integer, decimal
  - 042 → integer, octal
  - 0x2A → integer, hexadecimal
  - 0.15 → floating point
  - .15 → floating point
  - 1.5e2 → floating point
  - "ciao" → string

Variables & Constants

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- A variable is a “container” to store data
  - i.e. it allows to interact with the memory
  - Read data
  - Write data
- In Python a variable is created when a value is assigned
  - a = 57
  - b = “ciao”
  - pi = 3.141

Variables & Constants

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- In order to create a variable then
  - We need to choose a name
    - Only letters, digits or "_"
    - Have to start with a letter
  - Mnemonic names are to be preferred
  - No need to indicate a specific type
    - You can use "type(<variable name>)" to detect data type
    - Data type can change during variables lifetime

Variables & Constants

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- Operator “=” is used to assign (store) data
  - a = 17
  - b = a
  - x, y, z = "Orange", "Banana", "Cherry" # multiple assignements
- The content of a variable can “vary”
  - c = 17
  - …
  - c = “ciao”

Expressions

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- Combinations of **operators** and **operands**
- Mathematical
  - “+”, “-”, “*”, /, “//” (quotient), “%” (remainder), and “**” (exponential)
- Relational
  - “>”, “<”, “>=”, “<=”, “==” (equal to), “!=” (not equal to)
  - Can be chained (i.e. 0 < a < 17)
- Logical
  - “and”, “or”, “not”

Incomplete list!

Try it yourself!

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- Ask user for a radius value of a circle
- Compute both the circumference and area of the circle
  - Hints: area $\longrightarrow \pi r^2$, circumference $\longrightarrow 2\pi r$, $\pi \longrightarrow 3.141$
- Print them

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Data Structures

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- Complex programs need to manage also “big” data or even more “complex” data than single numbers or strings
- We are going to briefly introduce Python “Collections”
  - **Lists** → ordered and changeable, duplicate members
  - **(Tuples)** → ordered and unchangeable, duplicate members
  - **(Sets)** → unordered, unchangeable, unindexed, no duplicate members
  - **Dictionaries** → ordered and changeable. no duplicate members

Data Structures: lists

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- Lists allow to store multiple “items” in a single variable
- Square brackets are used to define them
  - `emptylist = []`
  - `workstation = ["mouse", "computer", "keyboard"]`
  - `print(workstation)`
- Lists can be heterogeneous
  - `workstation.append(42) # appending`
  - `workstation.extend(["RAM", "monitor"]) # extending`

Data Structures: lists

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- In a list single elements feature an “index”
- First element ⟶ 0
- We can access single elements using the index or even portions of the list

```
numbers = [1.414, 3.142, 2.718, 1.618, 1.202]
print(numbers[0])       # 1.414
print(numbers[0:2])     # [1.414, 3.142]
print(numbers[2:])      # [2.718, 1.618, 1.202]
print(len(numbers))     # 5
```

Try it yourself!

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- Define the following list
- thislist = ["apple", "banana", "cherry", "orange", "kiwi", "mango"]
  - Print the third element
  - Change the third element as “pear”
  - Using “.sort” sort the list
  - Using “.append” add “lemon” to the list
  - Using “.remove” remove “kiwi” from the list
  - Using “.pop” remove the fourth element from the list

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Data Structures: dictionaries

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- Dictionaries extend the lists allowing also to set the index
- Data are stored in a **key:value** fashion

```
person = {
    "name":     "Mario",
    "surname":  "Rossi",
    "IDs":      ["CI: AC2345Y", "PAT: 12349012"]
}
print(person)
```

Data Structures: dictionaries

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- The value of a single element can be accessed through the key
  - The key can be any **uniq** immutable object, i.e. strings, numbers, tuples
  - Values can be, as in the lists case, heterogeneous

  - `print(person["name"])`
  - `print(person.get("name"))`
  - `print(person.keys())`
  - `print(person.values())`
  - `print(person.items())`

Try it yourself!

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- Use the following Python dictionary that contains phone numbers
- tel = {"Bertozzi": 5845, "Rossi": 5722, "Lab": 5723, "Ugo": 3456}
  - Print the phone number of Lab using ".get()"
  - Remove the phone number of Ugo using ".pop()"
  - Add a phone number
  - Remove all elements using ".clear()"

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Control flow

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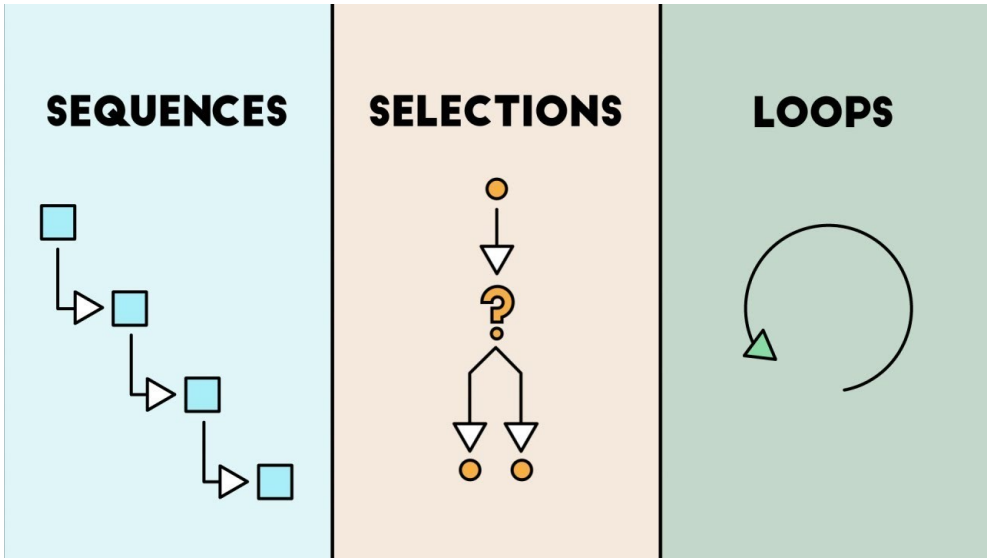
- Instructions are executed in the order they are met
- This is definitely limiting, we need to "control" the "flow" of execution



Indentation

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- Most programming languages do not care about indentation
- Humans tend to group similar things together
- Therefore indentation can be used to beautify your code
- But **for Python indentation is mandatory!**

```
# Python code
if foo:
    if bar:
        baz(foo, bar)
    else:
        qux()
```

Text

Control flow: selection

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- It allows to perform a different processing depending on evaluating a condition

```
temp = int(input("Give me a temperature: "))
if (temp >= 100):
    print("Put pasta in the water!”)
elif ( temp > 90):
    print("A bit early but just few minutes")
else:
    print("Do not be so impatient!")
```

Try it yourself!

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- Ask user an integer number
- Then print if the number is even or odd

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Control flow: Loops

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- Python provides two main possibilities for loops
- **for** loop
  - Typically used for a “fixed” amount of iterations
- **while** loop
  - Typically used until a specific condition occurs

Control flow: for

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- Used for iterating over a sequence
  - And to execute something each iteration

```
for x in range(10): # 0-9
    print(x)

fruits = ["Apple", "Orange"]
for fruit in fruits:
    print(fruit)
```

Control flow: for

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- For iterating over a dictionary we can use:

```
provinces = { "PR": "Parma", "MI": "Milano", …}
for acronym,province in provinces.items():
    print(acronym, province)
else:
   print("No more provinces!")
```

Control flow: while

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- Execute statements as long as a condition is true

```
i = 1
while i < 6:
    print(i)
    i += 1 # += increment and assign values at once
else:      # optional part
    print("Now the count reached:", i)
```

Try it yourself!

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- Use a *for* and a nested *if* to print odd numbers up to 99
- Rewrite previous code using a *while* loop
  - Is the *if* still necessary?

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Functions or Subroutines

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- Building blocks that ease a program development
  - Sub-problems matching
  - Easier to debug
  - Define once, use every time you need

Functions or Subroutines

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- The “def” keyword is used to define a function

```
def hello():
    print("Hello World!")

# to call a function, simply use function name + ()
hello()
# you can do this as many times as you need
hello()
```

Functions: arguments

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- Data can be passed into functions as arguments
- Or returned using the “return” or “yield” instructions

```
# Positional
def add(x, y):
    return x + y
# Keyword
def shout(phrase='Help me!'):
    print(phrase)
# Default
def echo(text, prefix = ''):
    print(prefix, text)
```

Functions: example

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- Compute Fibonacci numbers up to n

```
def fib(n):
    results = []
    a, b = 0, 1
    while a < n:
        results.append(a)
        a, b = b, a + b
    return results
```

Functions: yield

- Continuous Fibonacci computation

```
def fib():
    a, b = 0, 1
    while True:
        yield a
        a, b = b, a + b

for value in fib():
    print(value)
```

Try it yourself!

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- Define a function that find the maximum between two numbers and return it
- Define a function that sum all the numbers in a list

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There is more...

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- Python has other features:
  - Classes → OOP programming
  - Imports → code isolation and re-use
  - Error handling → exceptions
  - Docstrings → support for documenting all modules
  - ...

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Quick introduction to Python